PHYSLLM · GENERATIONS G0 → N02 → N03

PhysLLM: Residual Dynamics at Model Scale

One measurement programme, three generations: a pilot, a frozen evaluation manifest, and MoE routing inside a 117B-parameter model — every number shipped with SHA-256 provenance.

> **Claim** — the residual stream of a transformer can be measured the same way at every scale, and the protocol (observables → manifest → gates → provenance) is what makes the numbers comparable across generations.

3
generations: G0 · N02 · N03

14.4k
text x block rows captured

36
blocks · full / sliding attention

128/4
experts per token (top-4)

G0
pilot on a Colab bundle

›

N02
frozen observables + manifest

›

N03
MoE routing-conditional

Artifacts: **physllm** · **physllmN02** · **PhysLLM_N03_Antigravity_Workspace**
All numbers are project measurements. Figures rendered from project artifacts; no project file was modified.

G0 · WEEK 1

Pilot: is the residual stream measurable?

First question: do layer-wise hidden-state norms and updates $\Delta h_l = h_{l+1} - h_l$ come out finite and correctly shaped?

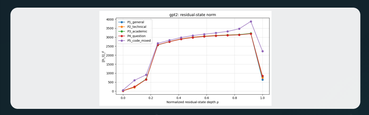


Fig. 1. G0 pilot output (Colab bundle, Qwen3 finalization v1.0.4).

N02 · PROTOCOL

Manifest before inference



Fig. 4. Candidate manifest v1.0.3 (preflight: 0 hard-check failures).

N03 · RESULT

Residual ratio falls, routing stays flat

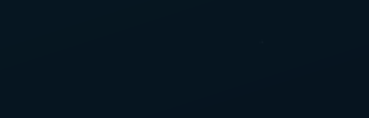


Fig. 6. Depth-band summary over 36 blocks; mean normalized entropy 0.968. Output/input ratio **1.351** → **1.234** → **1.137** across early/middle/late blocks, while routing entropy stays near uniform (H/ln4 ≈ 0.95–0.98).

N03 · STATUS

Done vs pending

Done: scope declaration, M1 capture script, gate tests, notebook simulator, real end-to-end capture on a tiny checkpoint (CPU, both transformers families), per-token routing artifact for 400 texts, T0 audit.

Pending: T0 option choice · the paid 80 GB session · repository approval.

N02 · WEEK 2

Three observables, frozen

Small HF causal LMs (**gpt2**, **pythia-160m**, 12 blocks) instrumented block by block, with the observable set **frozen for traceability**:

1. length-adjusted update norm $\|h_{l+1} - h_l\|_F/\sqrt{T}$
2. input–update cosine $\cos(h_l, \Delta h_l)$
3. output/input norm ratio

N02 · HONESTY

What the pilot is not

- The five pilot prompts are **excluded** from the main evaluation.
- The earlier synthetic chart was **not real model data**.
- N02's scientific hypothesis status stays **INCONCLUSIVE**; the baseline is frozen for traceability, not for claims.
- 24 code texts carry no domain word; 12 share one template.
- **frozen_for_confirmatory_analysis: false** — human review still pending.

N03 · INFRASTRUCTURE

80 GB is a hard wall

MXFP4 weights ≈ **60.8 GB**; 8K context ≈ 65.8 GB, 32K ≈ 68.6 GB.

Minimum: **one** 80 GB GPU (A100 80GB / H100 80GB / MI300X). L4 24GB, T4 16GB, A100 40GB are insufficient. Runtime: Triton ≥ 3.4, compute capability ≥ 7.5.

Development uses **gpt-oss-20b** (~16 GB) only for pipeline wiring.

ACROSS GENERATIONS

What carries forward

One observable definition survives every generation:

$$\Delta h_l = h_{l+1} - h_l \quad \cdot \quad \|\Delta h_l\|_F/\sqrt{T} \quad \cdot \quad \cos(h_l, \Delta h_l)$$

G0 asked whether it is measurable, N02 froze **what** to measure and **how** to sample, N03 reuses both and adds the router — with the same gates, the same SHA-256 discipline and the same habit of stating the limits next to the number.

N02 · RESULT

Cosine flips sign with depth

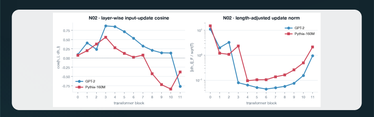


Fig. 2. N02 layer-wise summary (5 pilot prompts per model — descriptive, n=5). Cosine rises, then **turns negative in the last blocks** (GPT-2 −0.769 at block 11; Pythia −0.840 at block 10).

N03 · WEEK 3

Three branches, one active

N02 deferred three architecture families; N03 registers them as candidates and runs **M1 (MoE routing-conditional)**. M2 (MLA latent attention) and M3 (SSM / hybrid) stay **design-stage only** — no claim may use them yet.



Fig. 5. Target: **openai/gpt-oss-120b** — MXFP4 weights, bf16 activations.

N03 · GATES

Provenance is enforced, not documented

Every capture must pass the dataset gates (**G1, M1–M5, P1**) and ships provenance:

- **sha** manifest SHA-256 · code SHA + git revision/dirty
- **meta** tokenizer ID · activation dtype · seed 0 · package versions
- **tests** 30 gate tests passing (incl. an independent routing recomputation)

Resume is idempotent and refuses to append into an out-dir captured from a different manifest.

N03 · RESULT

Attention vs MLP branch

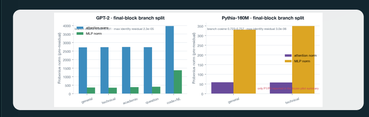


Fig. 3. Final-block branch split; identity residual ≤ 2.3e-5 (GPT-2). Attention dominates GPT-2 (cosine 0.37–0.43) while Pythia is MLP-dominant (0.72–0.76) — the same observable, two architectures.

N03 · OBSERVABLES

Routing, not just residuals

Residual stream measures are inherited from N02; N03 adds the branch split and the router:

- **attn_share** $\text{attn}^2/(\text{attn}^2+\text{moe}^2)$ of the pre-residual output
- **routing** top-4 expert ids + gate weights (softmax over the **top-k logits**, not the full 128)
- **entropy** routing entropy and H/ln k (k = 4)
- **load** per-text/layer expert load histogram

N03 · AUDIT

T0: three texts out of band

The tokenizer audit recomputes token counts for every candidate text. **gpt2** and **pythia-160m** come back clean (0 mismatches).

gpt-oss leaves 3 long-tier texts out of band (127–128 tokens against a 129 floor) — and they are not random: 11 of 33 long × question, dialogue texts now sit within 2 tokens of the floor, while other long styles clear it by 21+ tokens.

Options A / B / B′ / C are quantified in the decision memo — **choice pending**.

LIMITS

Stated next to the numbers

- Exploratory **descriptive** statistics — no causal or topic-clustered inference.
- No independent validation against the model's real router dispatch or an fp32 reference run.
- Candidate manifest **not frozen** for confirmatory analysis.
- N02 pilot is 5 prompts per model; N03 blocks are a capture, not a behavioural claim.
- Max relative branch-identity residual 0.0187 (N03) · 2.3e-5 (N02).